

PCB Ordering Instructions — Personalized Weather Station

Overview

- The Personalized Weather Station PCB was designed in KiCad.
- The board is manufacturing-ready and can be fabricated through:
 - JLCPCB
 - PCBWay
 - OSHPark

Files Required for Manufacturing

- Gerber ZIP folder
- Drill file
- Bill of Materials (BOM)

Recommended PCB Specifications

- Board Material: FR4
- Layer Count:
 - 2-layer version
 - 4-layer experimental version
- Board Thickness: 1.6mm
- Copper Weight: 1 oz
- Surface Finish: HASL Lead-Free
- Solder Mask: Green
- Silkscreen: White

USB-C Integration

The PCB currently includes two USB-C connectors:

1. USB-C Power Input

- Dedicated to powering the board
- Intended to simplify external power delivery
- Replaces older screw terminal-only power concepts

2. USB-C Communication Port

- Dedicated to communication between the PCB and external computer software
- Intended for USB serial data transfer
- Allows environmental data to be processed and visualized through a Python-based computer application

Note:

- USB-C implementation is still considered a work-in-progress during this stage of development and testing.

Manufacturing Notes

- ERC verification completed

- DRC verification completed
- Standard PCB manufacturing tolerances are acceptable
- No impedance-controlled routing required
- All components selected using commercially available parts

Assembly Notes

Carefully verify orientation and placement of:

- ATmega328P-A microcontroller
- BME280 environmental sensor
- Crystal oscillator
- USB-C connectors
- LED polarity
- Capacitors

Firmware Programming

- Firmware is flashed through ISP Header (J2)
- Compatible programming methods:
 - USBasp programmer
 - Arduino configured as ISP

Power Input

The board may currently be powered through:

- Dedicated USB-C power connector
- External screw terminal power input (if populated)

Important:

- Do not connect multiple power sources simultaneously

Post-Assembly Validation

After assembly:

- Verify no shorts exist between VCC and GND
- Verify proper USB-C connector soldering
- Verify component orientation
- Apply power and confirm the board powers correctly
- Flash firmware to the ATmega328P-A
- Verify environmental sensor readings and USB communication functionality

Final Status

- PCB is manufacturing-ready
- Fabrication files are complete
- USB-C communication and power integration are currently in development and testing
- Design prepared for future expansion and software integration